The Singular Value Decomposition (SVD)

Both diagonalization and QR factorization have wide application in applied settings in conditions when numerical error or computational cost is an issue. Another decomposition of a matrix, the singular value decomposition, is also useful and informative.

Let A be a $n \times m$ matrix with then there exist matricies

$U(\text{ orthogonal an } n \times n)$,
$V(\text{ orghogonal and } m \times m)$, and
$\Sigma = \begin{bmatrix} D_{m\times m} \\ 0_{(n-m)\times m} \end{bmatrix}$ if $n \geq m$ or $\Sigma = \begin{bmatrix} D_{m\times m} & 0_{n\times(m-n)} \end{bmatrix}$ if $m > n$ note Σ is n × m

Such that $A = U\Sigma V^T$

Where $D_{ij} = \sigma_i \delta_{ij}$ and $\sigma_i = \sqrt{\lambda_i}$ which are called singular values of A and are the square roots of the positive eigen values of A^TA or AA^T, which are the same. Usually ordered from greatest to least down the diagonal.

$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$

$\sigma_1 > \sigma_2 > \sigma_3 \cdots$

The columns of U are the orthogonal and normalized eigen vectors of AA^T which are called the left singular vectors of A ordered to be paried up with σ_i

The columns of V are the orghogonal and normalized eigen vectors of A^TA which are called the right singular vectors of A ordered to be paried up with σ_i

Example: Find the singular value decomposition of $A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}$ (n m, 3×2)

$AA^T = \begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & 0 & 4 \end{bmatrix}$ (3×2, 2×3)

$A^TA = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}$ (2×3, 3×2)

$\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{bmatrix}$

Find eigen values A^TA

$\begin{vmatrix} 5-\lambda & 2 \\ 2 & 8-\lambda \end{vmatrix} = (5-\lambda)(8-\lambda) - 4$

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Finding V

$$\begin{vmatrix} 5-\lambda & 2 \\ 2 & 8-\lambda \end{vmatrix} = (5-\lambda)(8-\lambda) - 4 = 0$$

$$= 40 - 8\lambda - 5\lambda + \lambda^2 - 4 = \lambda^2 - 13\lambda + 36 = 0$$

$$(\lambda - 9)(\lambda - 4) = 0$$

$$\lambda_1 = 9 \quad \lambda_2 = 4$$

$$\sigma_1 = \sqrt{9} = 3 \quad \sigma_2 = \sqrt{4} = 2$$

$$\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 5 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & 0 & 4 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 5-9 & 2 \\ 2 & 8-9 \end{bmatrix} \qquad \begin{bmatrix} 5-4 & 2 \\ 2 & 8-4 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$\frac{1}{2}R_1 + R_2$

$$\begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix}$$

$-2R_1 + R_2$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

$x_2 = t$

$-4x_1 + 2t = 0$

$-4x_1 = -2t$

$x_1 = -\frac{1}{2}t$

$$\begin{bmatrix} -1/2 \\ 1 \end{bmatrix} t$$

$x_2 = t$

$x_1 + 2x_2 = 0$

$x_1 = -2t$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix} t$$

$$\vec{v}_1 = \frac{1}{\sqrt{(-\frac{1}{2})^2 + 1}} \begin{bmatrix} +1/2 \\ 1 \end{bmatrix} = \frac{2}{\sqrt{5}} \begin{bmatrix} +1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} +1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\frac{1}{\sqrt{(-2)^2 + 1^2}} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{matrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{matrix}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

The Singular Value Decomposition (SVD)

finding U

$$AA^T = \begin{bmatrix} 5 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & 0 & 4 \end{bmatrix} \rightarrow \det(A-\lambda I) = \begin{vmatrix} 5-\lambda & 2 & 4 \\ 2 & 4-\lambda & 0 \\ 4 & 0 & 4-\lambda \end{vmatrix}$$

I'll expand across the bottom row

$$= 4\begin{vmatrix} 2 & 4 \\ 4-\lambda & 0 \end{vmatrix} - 0\begin{vmatrix} 5-\lambda & 4 \\ 2 & 0 \end{vmatrix} + (4-\lambda)\begin{vmatrix} 5-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix}$$

$$= 4(2(0) - (4-\lambda)4) - 0((5-\lambda)(0) - 2(4)) + (4-\lambda)((5-\lambda)(4-\lambda) - 2(2))$$

$$= -16(4-\lambda) + (4-\lambda)(5-\lambda)(4-\lambda) - 4(4-\lambda)$$

$$= (4-\lambda)(-16 + (5-\lambda)(4-\lambda) - 4)$$

$$= (4-\lambda)(-20 + 20 - 4\lambda - 5\lambda + \lambda^2)$$

$$= (4-\lambda)(\lambda^2 - 9\lambda) = \lambda(4-\lambda)(\lambda-9)$$

$$\Rightarrow \lambda_1 = 9 \quad \lambda_2 = 4 \quad \lambda_3 = 0$$

Now we find eigen vectors...

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The Singular Value Decomposition (SVD)

Finding U

$AA^T = \begin{bmatrix} 5 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & 0 & 4 \end{bmatrix}$ $\quad \lambda_1 = 9, \lambda_2 = 4, \lambda_3 = 0$

for $\lambda_1 = 9$

$$\text{Null}\left(\begin{bmatrix} 5-9 & 2 & 4 \\ 2 & 4-9 & 0 \\ 4 & 0 & 4-9 \end{bmatrix}\right)$$

$$\text{Null}\left(\begin{bmatrix} -4 & 2 & 4 \\ 2 & -5 & 0 \\ 4 & 0 & -5 \end{bmatrix}\right) \to \left[\begin{array}{ccc|c} -4 & 2 & 4 & 0 \\ 2 & -5 & 0 & 0 \\ 4 & 0 & -5 & 0 \end{array}\right]$$

$R_1 + R_3$

$$\begin{array}{r} -4\ 2\ 4\ 0 \\ +\ 4\ 0\ -5\ 0 \\ \hline 0\ 2\ -1\ 0 \end{array}$$

$-2R_2 + R_3$

$$\begin{array}{r} -4\ 10\ 0\ 0 \\ 4\ 0\ -5\ 0 \\ \hline 0\ 10\ -5\ 0 \end{array}$$

$$\left[\begin{array}{ccc|c} -4 & 2 & 4 & 0 \\ 0 & 10 & -5 & 0 \\ 0 & 2 & -1 & 0 \end{array}\right]$$

$-5R_3 + R_2$

$$\begin{array}{r} 0\ -10\ +5\ 0 \\ 0\ 10\ -5\ 0 \\ \hline 0\ 0\ 0\ 0 \end{array}$$

$$\left[\begin{array}{ccc|c} -4 & 2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} -4 & 2 & 4 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$-4x_1 + 2x_2 + 4x_3 = 0$, $2x_2 - x_3 = 0 \to x_3 = t$

$2x_2 - t = 0 \quad x_2 = \frac{1}{2}t$

$-4x_1 + 2\left(\frac{1}{2}t\right) + 4t = 0$

$-4x_1 + 5t = 0$

$\Rightarrow x_1 = \frac{5}{4}t$ $\quad \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5/4 \\ 1/2 \\ 1 \end{bmatrix} t$

Normalize $\dfrac{1}{\sqrt{(5/4)^2 + (1/2)^2 + 1^2}} \begin{bmatrix} 5/4 \\ 1/2 \\ 1 \end{bmatrix} = \dfrac{1}{\sqrt{\frac{25}{16} + \frac{4}{16} + \frac{16}{16}}} \begin{bmatrix} 5/4 \\ 1/2 \\ 1 \end{bmatrix} = \dfrac{1}{\sqrt{45}} \begin{bmatrix} 5/4 \\ 1/2 \\ 1 \end{bmatrix} = \dfrac{4}{\sqrt{45}} \begin{bmatrix} 5/4 \\ 1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5/\sqrt{45} \\ 2/\sqrt{45} \\ 4/\sqrt{45} \end{bmatrix}$

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for $\lambda_2 = 4$

$$\text{Null}\left(\begin{bmatrix} 5-4 & 2 & 4 \\ 2 & 4-4 & 0 \\ 4 & 0 & 4-4 \end{bmatrix}\right)$$

$$\text{Null}\left(\begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix}\right) \to \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 2 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{array}\right]$$

$-2R_2 + R_1$

$$\begin{array}{r} -4\ 0\ 0 \\ 4\ 0\ 0 \\ \hline 0\ 0\ 0 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_1 + 2x_2 + 4x_3 = 0$

$2x_1 = 0 \Rightarrow x_1 = 0$

$\to x_3 = t$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2t \\ t \end{bmatrix} \to \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Normalize $\dfrac{1}{\sqrt{(-2)^2 + 1^2}} \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} = \dfrac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$

$$= \begin{bmatrix} 0 \\ -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

The Singular Value Decomposition (SVD)

for $\lambda_3 = 0$ $\quad \text{Null}\left(\begin{bmatrix} 5 & 2 & 4 \\ 2 & 4 & 0 \\ 4 & 0 & 4 \end{bmatrix}\right)$

$$\left[\begin{array}{ccc|c} 5 & 2 & 4 & 0 \\ 2 & 4 & 0 & 0 \\ 4 & 0 & 4 & 0 \end{array}\right] \quad \begin{matrix} -2R_2+R_3 \\ -4 \;\; -8 \;\; 0 \;\; 0 \\ 4 \;\; 0 \;\; 4 \;\; 0 \\ \hline 0 \;\; -8 \;\; 4 \;\; 0 \end{matrix} \quad \begin{matrix} -4R_1+5R_2 \\ -20 \;\; -8 \;\; -16 \;\; 0 \\ 20 \;\; 0 \;\; 20 \;\; 0 \\ \hline 0 \;\; -8 \;\; 4 \;\; 0 \end{matrix}$$

$$\left[\begin{array}{ccc|c} 5 & 2 & 4 & 0 \\ 0 & -8 & 4 & 0 \\ 0 & -8 & 4 & 0 \end{array}\right] \quad R_2+R_3$$

$$\left[\begin{array}{ccc|c} 5 & 2 & 4 & 0 \\ 0 & -8 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{matrix} 5x_1+2x_2+4x_3=0 \\ -8x_2+4x_3=0 \\ \rightarrow x_3 = t \end{matrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t \\ \frac{1}{2}t \\ t \end{bmatrix} = \begin{bmatrix} -1 \\ \frac{1}{2} \\ 1 \end{bmatrix} t$$

Normalizing $\dfrac{1}{\sqrt{(-1)^2+(\frac{1}{2})^2+(1)^2}} = \dfrac{1}{\sqrt{1+\frac{1}{4}+1}} = \dfrac{1}{\sqrt{\frac{4}{4}+\frac{1}{4}+\frac{4}{4}}} = \dfrac{1}{\sqrt{\frac{9}{4}}} = \frac{1}{3/2}$

$= 2/3$ so $\dfrac{2}{3}\begin{bmatrix} -1 \\ \frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$

So we have

$$\vec{v}_1 = \begin{bmatrix} 5/\sqrt{45} \\ 2/\sqrt{45} \\ 4/\sqrt{45} \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 0 \\ -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \quad \vec{v}_3 = \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}$$

So

$$U = \begin{bmatrix} 5/\sqrt{45} & 0 & -2/3 \\ 2/\sqrt{45} & -2/\sqrt{5} & 1/3 \\ 4/\sqrt{45} & 1/\sqrt{5} & 2/3 \end{bmatrix} \quad V = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix} \quad \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$$

The Singular Value Decomposition (SVD)

```
octave:71> A=[1 2; 2 0; 0 2]
A =

   1   2
   2   0
   0   2

octave:72> [U1, DU1]=eig(A*A')
U1 =

   2/3        0   963/1292
  -1/3  -305/341   963/3230
  -2/3  1292/2889  963/1615

DU1 =

Diagonal Matrix

   0   0   0
   0   4   0
   0   0   9

octave:73> Sigma=[3 0; 0 2; 0 0]
Sigma =

   3   0
   0   2
   0   0

octave:74> U=[U1(:,3) U1(:,2) U1(:,1)]
U =

   963/1292          0       2/3
   963/3230   -305/341      -1/3
   963/1615  1292/2889      -2/3

octave:75> [V1, DV1]=eig(A'*A)
V1 =

  -305/341  1292/2889
  1292/2889  5473/6119

DV1 =

Diagonal Matrix

   4   0
   0   9

octave:76> V=[V1(:,2) V1(:,1)]
V =

  1292/2889   -305/341
  5473/6119  1292/2889

octave:77> U*Sigma*V'
ans =

   1   2
   2   0
   0   2
```

$[U, S, V] = svd(A)$

The Singular Value Decomposition (SVD)

Applications of the SVD



562 × 562

Image Compression: We can think of the image above as a matrix where we assign a gray scale intensity to each pixel position. So we have a matrix of numbers between 0 and 255.

If you recall, in the matrix Σ is ordered with the largest singular values first ang going down from there. Our SVD product can be written in the form

$$A = U\Sigma V^T = \sigma_1\vec{u}_1\vec{v}_1^T + \sigma_2\vec{u}_2\vec{v}_2^T + \cdots + \sigma_n\vec{u}_n\vec{v}_n^T$$

$\vec{u}_i$ is $n \times 1$ $\vec{v}_i^T$ is $1 \times m$ so $\vec{u}_i\vec{v}_i$ is $n \times m$ the same as A

From this we might note that if the smallest singular values are very small, they don't contribute much information to A. So what would happen to the image if we throw away some of terms being scaled by small singular values?

So we might say: $A \approx \sigma_1\vec{u}_1\vec{v}_1^T + \sigma_2\vec{u}_2\vec{v}_2^T + \cdots + \sigma_k\vec{u}_k\vec{v}_k^T$

Where $k < n$ keeping in mind that $\sigma_1 > \sigma_2 > \sigma_3 > \cdots > \sigma_n$

In matlab matrix multiplication notation keeping the first 10 terms would read

U(:,1:10)S(1:10,1:10)V(:,1:10)'

The Singular Value Decomposition (SVD)



Keeping only 10 singular values

U(:,1:10)S(1:10,1:10),V(:,1:10)'



Keeping only 50 singular values

U(:,1:50)S(1:50,1:50),V(:,1:50)'



Keeping only 100 singular values

U(:,1:100)S(1:100,1:100),V(:,1:100)'



Original 562X562

The Singular Value Decomposition (SVD)

The condition number of a matrix

The condition number of a matrix A is $\kappa(A) = \frac{\sigma_{max}}{\sigma_{min}}$ This number gives a sense of how terrible an inversion of a matrix might be!

```
>> B=randi([-10 10], 5 ,3)

B =

     -2     -4     -8
      1     -1     -3
     -2    -10     -6
      3     10      0
      3     -7     -3

>> A=[B B(:,2)-0.001, B(:,1)-.01]

A =

     -2     -4     -8     -4001/1000     -201/100
      1     -1     -3     -1001/1000       99/100
     -2    -10     -6    -10001/1000     -201/100
      3     10      0      9999/1000      299/100
      3     -7     -3     -7001/1000      299/100

>> [U,S,V]=svd(A)

U =

   -431/1274     -759/1090      385/1053      569/1102      302/15201
   -427/4964     -421/1285     -229/1069     -466/1487     -588/683
  -1139/1791     -231/1943      239/1902     -268/395       306/943
   1171/2076    -3543/5656    -1316/4877     -339/1138      604/1689
   -941/2386     -356/10045    -2995/3501      547/1864      305/1919


S =

   8644/353          0          0          0          0
          0   4762/609          0          0          0
          0          0   2905/442          0          0
          0          0          0   59/17298          0
          0          0          0          0          *


V =

    295/3046     -243/2783     -997/1433     -272/387      -221/3141
    563/851      -415/1893      269/2252     -538/7655      489/695
   1107/3404     2049/2173     -511/7148      -30/19169     *
    698/1055     -966/4411      295/2467      188/2669     -489/695
    265/2726    -1028/12093    -2441/3515     2785/3954      221/3141

>> svd(A)

ans =

   8644/353
   4762/609
   2905/442
     59/17298
   1/5614388000771653
```

This condition number is huge!
We would expect any inversion to be horrible!

```
>> S(1,1)/S(5,5)

ans =

   137480924864676784
```

The Singular Value Decomposition (SVD)

```
>> x=[1;2;3;4;5]

x =

     1
     2
     3
     4
     5

>> b=A*x

b =

  -60.0540
   -9.0540
  -90.0540
   77.9460
  -33.0540

>> inv(A)*b
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND =  3.663224e-18.

ans =

    1.0396
    3.5995
    3.0000
   12.4005
    3.4275
```

$x = A^{-1}b$

$x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}$

Why do we have this problem? Let's think about inverting the SVD version of A....

$A\vec{x} = \vec{b}$

$\vec{x} = A^{-1}\vec{b}$

$(USV^T)\vec{x} = \vec{b}$

$USV^T x = b$ $\quad U^T = U^{-1}$

$I - U^T U S V^T \vec{x} = U^T b$

$S V^T \vec{x} = U^T b$

$S^{-1} S V^T \vec{x} = S^{-1} U^T b$

$I \quad V^T \vec{x} = S^{-1} U^T b \rightarrow \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$

$S^{-1} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1/\sigma_1 & & \\ & \ddots & \\ & & 1/\sigma_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$

$\begin{bmatrix} 1/\sigma_1 y_1 \\ 1/\sigma_2 y_2 \\ \vdots \\ 1/\sigma_n y_n \end{bmatrix} \rightarrow 1/\sigma_n y_n$ numerical error

$S = \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_2 & \\ & & \ddots \\ 0 & & \sigma_n \end{bmatrix}$ $\sigma_n \approx 0$

$S^{-1} = \begin{bmatrix} 1/\sigma_1 & 1/\sigma_2 & \cdots & 1/\sigma_n \end{bmatrix}$ → HUGE